

CS383 - Software Engineering

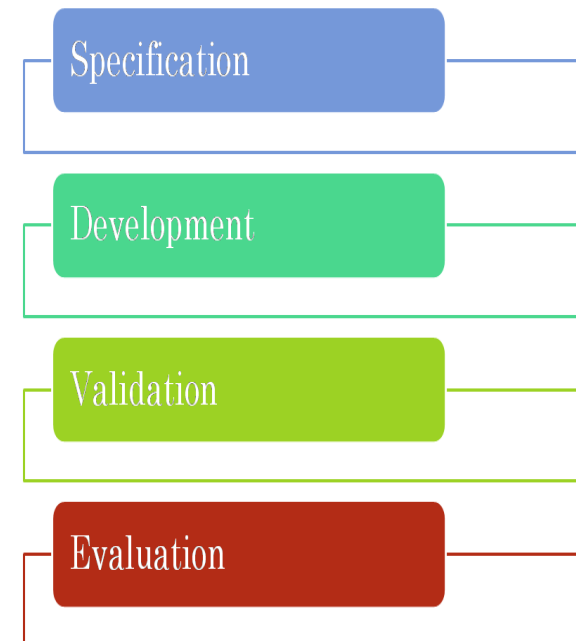
Introduction (II)

Topics to be covered in this lecture

- Software Engineering activities
- SE and the web
- Software as a Service
- SE ethics

Software engineering and process activities

- Software specification: define the software that is to be produced and the constraints on its operation.
- Software development: design and implement the software according to specifications
- Software verification and validation: the software is checked to ensure that it is what the customer needs
- Software evolution: the software is modified to reflect changing customer and to evolve to address market requirements/challenge.



Software engineering and the Web

- In the last two decades, web-based systems has be developed and accessed using web browsers
- That led to the development of a vast range of new system products
 - Product deliver services accessed over the Web
- Web-browsers also evolved from running small programs to enterprise and organizational software
 - Nowadays software don't need to be installed locally
 - Web-based systems are accessible worldwide instantly
 - Many software now migrated to use the web

Software engineering and the Web (cont.)

- Software (on the web) helped (from software developer perspective)
 - Faster software delivery
 - Cheaper to changed and upgrade
 - Reduce cost

Software engineering and the Web (cont.)

Because of fast evolvement of software organization and web-based systems

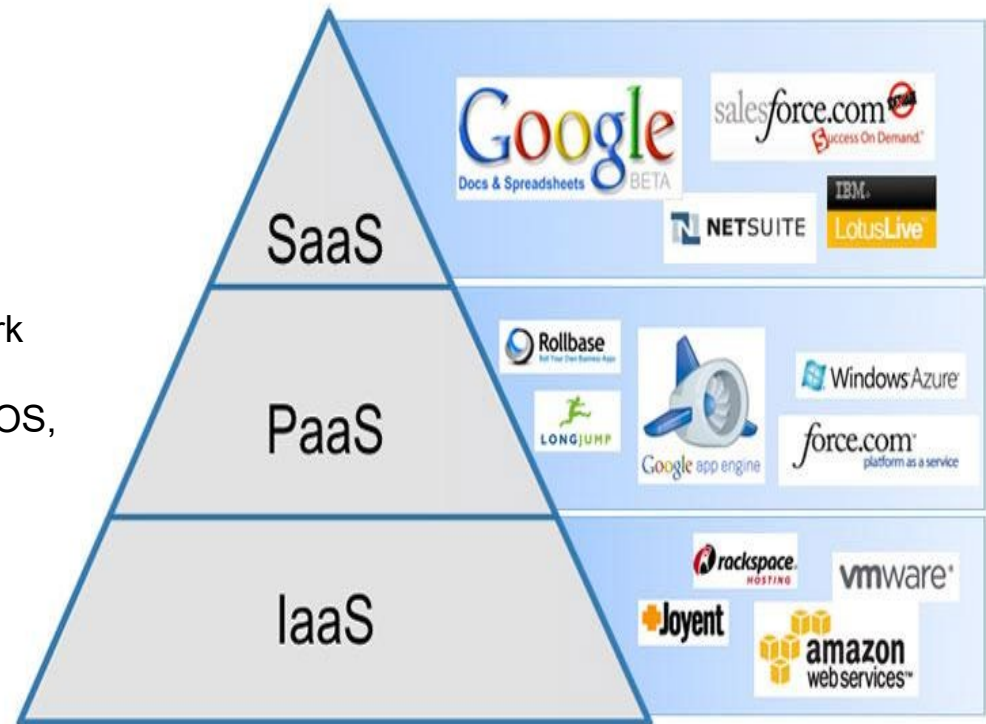
- Software reuse has rapidly become an asset
- Specifying all requirements in advance is not practical anymore
 - Use incremental development-based methods to deliver software
- However, user interfaces are *constrained* by the capabilities of web browser

Software as a Service (SaaS)

- Instead of a complete software installed locally, now let us have it as a service
- Individual service (software) will be run on a computing cloud on the web

Platform: provide cloud components and framework for software creation.
Eliminate concerns about OS, storage or infrastructure

Infrastructure: provide scalable resources like servers, network, storage, OS via virtualized technology



Software as a Service (SaaS)

- **Benefits:**

- Pay as you go (monthly, annually)
- Accessible anywhere
- No need for expensive computers

- **Drawbacks:**

- Lack of control
- Vulnerable: security and trust
- Limited range of applications
- Internet dependent
- Performance issues